



BMW's futuristic scooter

Motorrad Concept Link is a zero-emission mobility-scooter concept presented by BMW.
<http://dgit.in/MotorradLink>



Nuclear Contingency Plan

Hawaii's Department of Defence is revamping its emergency plans for nuclear missile attacks.
<http://dgit.in/NukeHawaii>

A few years later, an English biologist named J.B.S. Haldane independently proposed similar ideas in a short article published in the *Rationalist Annual*. Oparin and Haldane thus helped shape the “primordial soup” model of abiogenesis where life could, in fact, emerge organically from the right chain of chemical reactions under suitable external conditions. As compelling as Oparin and Haldane's hypotheses were, there was no proof.

RECREATING LIFE IN THE LAB

What are the fundamental blocks that come together to create life? If we know what they are, can we recreate life in the laboratory? You would think that post the primordial soup model, things would become simpler to explain. But you would be very wrong.

Harold Urey won the 1934 Nobel Prize in Chemistry for his pioneering work on isotopes and later helped build the atomic bomb. Post the World War II period, he became interested in the chemistry of outer space during the formation of the solar system. One day while delivering a lecture, he spoke about how lack of oxygen on Earth offered the perfect conditions for life in the models presented by Oparin and Haldane – the fragile chemicals would have been destroyed if they came in contact with oxygen.

In the audience was doctoral student Stanley Miller, who approached Urey and presented an experiment intended to recreate the environment of early Earth. Conducted in 1952, this later famously became known as the Miller-Urey experiment. Miller mixed methane, ammonia, and hydrogen in a flask, then passed water vapour through this while simulating lightning strikes using an electrode. After a day, the solution had turned pink, and continued to turn into a deeper shade of red as the experiment went on for a week. When Miller analysed the solution, he was able to identify two amino acids: Glycine and alanine. This was huge.

Amino acids are considered the building blocks of life. By demonstrat-

Eureka Moment



“In 2005, I had been struggling for weeks with an energetic chicken-and-egg problem. When I realised

that geochemical synthesis of methyl groups would solve that problem, things made sense quickly. It seemed almost illogical at the time, but today we know that hydrothermal systems generate methane, meaning that the methyl groups have to be there, too.”

— DR. WILLIAM F. MARTIN
 INSTITUTE OF MOLECULAR EVOLUTION,
 UNIVERSITY OF DÜSSELDORF



“The insight that most changed my approach to the origin of life was simply that we had to think about the

origins of cells and not simply the origins of separate components such as membranes, genetic materials, and peptides. Ever since, the assembly of primitive cells that grow, divide and evolve has been the focus of research in my laboratory.”

— DR. JACK W. SZOSTAK
 WINNER OF 2009 NOBEL PRIZE FOR
 PHYSIOLOGY OR MEDICINE, PROFESSOR
 OF CHEMISTRY AND CHEMICAL BIOLOGY,
 HARVARD UNIVERSITY

ing that you could create organic molecules from inorganic matter given the right conditions, the Miller-Urey experiment secured its place as a breakthrough in abiogenesis research. However, there was a problem, it still offered no useful insights into how actual living beings may have evolved from those simple amino acids, and that's where the real complexity was.

Biologically, a cell is the basic unit of life – you can examine anything from amoeba to human tissue under a microscope and you will see the same

thing – self-contained entities that are capable of harnessing energy and self-replication, each made up of complex molecular structures called *organelles*. It was hard to conceive how amino acids could huddle together to make up all that stuff. A link was found but the big picture was still missing pieces.

ORIGIN THEORIES

Abiogenesis is replete with theories, some of them contradictory, others plain ridiculous, scientists are known to form tribes and even polarise journalists against the “opponents” by calling their research stupid. Here, we look at the most plausible ones.

Have you ever thought about how difficult it is to pin down what being alive means? Is it the presence of hydrocarbons? Well, sugar has hydrocarbons, but it's not really alive. Is it the process of consuming energy and creating excrement? Fire fits the description, it certainly isn't alive in any way we think. Is it having a soul? Whatever that means? Not really, now we're stepping out of science and going into metaphysics and folklore. So then what it is?

Most scientists now agree on a few essential features that define what is living: The ability to harness energy for sustenance, a mechanism for self-replication, and means of self-preservation. Most proposed models of abiogenesis take one of those features and try to build a valid theory by trying to fit in the missing parts.

While Stanley Miller was trying to create organic molecules from non-living matter, Alfred Hershey and James Crick were trying to crack the code of life at the University of Cambridge in UK. In 1952, they proposed the double-helix structure of DNA, or deoxyribonucleic acid – the molecule that carried genetic information in living organisms. DNA informs cells how to synthesise proteins, but DNA is too precious and is therefore guarded closely within the nucleus, another molecule called RNA acts as the messenger between DNA and protein. A British chemist named Leslie Orgel first proposed that